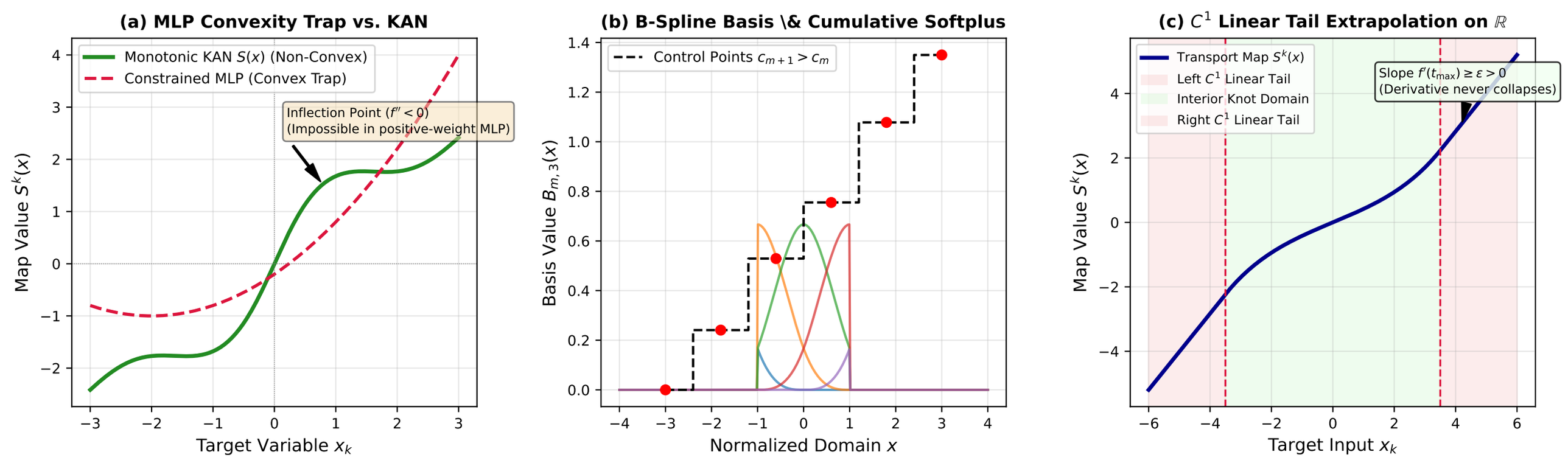
(a) MLP Convexity Trap vs. KAN
Monotonic KAN $S(x)$ (Non-Convex)
Constrained MLP (Convex Trap)
Inflection Point ($f'' < 0$)
(Impossible in positive-weight MLP)
Map Value $S^k(x)$
Target Variable x_k
(b) B-Spline Basis \& Cumulative Softplus
Control Points $c_{m+1} > c_m$
Basis Value $B_{m,3}(x)$
Normalized Domain x
(c) C^1 Linear Tail Extrapolation on $\mathbb{R}$
Transport Map $S^k(x)$
Left C^1 Linear Tail
Interior Knot Domain
Right C^1 Linear Tail
Slope $f'(t_{\max}) \geq \varepsilon > 0$
(Derivative never collapses)
Map Value $S^k(x)$
Target Input x_k